GIS APPLICATION IN TELECOM

5-Day Intensive Training Program

DAY 1 LAB MANUAL

GIS concepts, coordinate systems, ArcGIS Pro, geodatabase, domains, subtypes, relationship classes, raster / DEM

Version 4.0 | Follows presentation *Day 1_ready.pptx* (also saved as *Day1_GIS_Application_in_Telecom_Ready.pptx*)

Platform: ArcGIS Pro | Duration: 1 day (09:00 – 16:00)

Document	Detail
Course	GIS Application in Telecom
Day	Day 1 — same topics as the presentation, in the same order
Version	4.0
Presentation	Day 1_ready.pptx
How to use	Instructor shows a slide block. Class then does the matching lab steps in ArcGIS Pro. No sample CSV or GeoJSON files are required.

NOTE: Do not load synthetic MetroTel tables for this day. You create empty shapefiles and geodatabase objects, type the key fields, digitize a few features by clicking, then build relationship classes. That is enough to match the presentation.

How this manual maps to the presentation

Presentation	Slides (approx.)	Do this in the manual
Title, 5-day agenda, Day 1 overview, Day 1 agenda, outcomes	1–6	Read only — no software
Part 1 — GIS concepts and data types	7–16	Lab 1 — open Pro and see map + table
Part 2 — Coordinate systems	17–24	Lab 2 — set the map CRS
Part 3 — ArcGIS Pro and the geodatabase	25–31	Lab 3 — shapefile, GDB, feature dataset, feature classes
Domains and subtypes	32–34	Lab 4 — coded domain, range domain, Towers subtypes

Relationship classes, ERD, cardinality, irrigation example	35–41	Lab 5 — keys then relationship classes (irrigation + telecom)
Part 4 — Raster analysis and DEM	42–50	Lab 6 — add a DEM and run Slope / Hillshade / Aspect
Instructor corrections (yellow text)	51	Instructor only — delete before class

1. Learning outcomes

Match presentation: Day 1 Learning Outcomes

By the end of Day 1 you will be able to:

- Explain GIS as a system (data, software, hardware, people, methods).
- Choose vector or raster for a given question (assets vs coverage / terrain).
- Distinguish GCS vs projected CRS, and explain datum and WGS84.
- Create a shapefile, a file geodatabase, a feature dataset, and feature classes in ArcGIS Pro.
- Create a domain and subtypes, and assign them to fields.
- Put a primary key on the parent and a foreign key on the child, then create a 1:M relationship class.
- Add a DEM and derive slope, aspect, and hillshade.

2. Day agenda

Time	Presentation block	Lab
09:00–09:30	Slides 1–6 — program, 5-day agenda, Day 1 agenda & outcomes	—
09:30–10:30	Part 1 — What GIS is, keywords, functions, components, vector/raster	Lab 1
10:30–10:45	Break	—
10:45–12:00	Part 2 — Earth, datum, WGS84, GCS vs PCS	Lab 2
12:00–13:00	Lunch	—
13:00–14:15	Part 3 — Pro, geodatabase, shapefile, feature class, feature dataset	Lab 3
14:15–14:50	Domains and subtypes	Lab 4
14:50–15:00	Break	—
15:00–15:40	ERD, cardinality, irrigation + telecom 1:M	Lab 5
15:40–16:00	Raster / DEM / “Let’s explore it in ArcGIS Pro”	Lab 6

3. One-time project setup

Do this once before Lab 1. All later labs use the same project.

1. Open **ArcGIS Pro**.
2. Click **New** → **Map**.
3. Name the project `Day1_GIS_Telecom`. Save it in a local folder you can find again (for example `C:\GIS_Training\Day1`).
4. When the map opens, look at four places you will use all day:
   - **Catalog** pane — data and the geodatabase
   - **Contents** pane — layers on the map
   - **Map** view — geography
   - **Ribbon** (Map, Edit, Analysis, View)
5. Catalog pane → right-click **Folders** → **Add Folder Connection** → select your Day 1 folder.
6. Catalog → right-click **Databases** → **New File Geodatabase** → name it `Day1.gdb`.
7. Map tab → **Basemap** → **Imagery** or **Topographic** (context only).

CHECKPOINT: You have a named project, a folder connection, and an empty `Day1.gdb`.

4. Part 1 — GIS concepts and data types

Presentation: Part 1 · slides "What is GIS" through "Advantages and Disadvantages of Raster"

4.1 What to take from the slides (short)

Slide idea	Remember this
What is GIS	A Geographic Information System captures, stores, analyzes, and presents geographically referenced data. Software alone is not a GIS — people, hardware, data, and methods belong too.
Keywords	**Geographic** = location on Earth. **Information** = attributes / tables. **Systems** = hardware, software, data, procedures working together.
Functions	Acquire, store, manage, analyze, visualize.
Components	Hardware, software (ArcGIS Pro), data (spatial + attribute), people, methods.
Vector	Points, lines, polygons with exact coordinates. Use for towers, fiber, parcels, homes.
Raster	A grid of cells. Use for elevation, imagery, rainfall, coverage surfaces.
Attributes / metadata	Attributes = the "what" attached to geometry. Metadata = source, date, CRS, accuracy.

4.2 Lab 1 — See map and table together

Objective: prove the GIS idea from the slides — one feature is a shape on the map and a row in a table.

1. On the ribbon, open the **Map** tab. Use **Explore** to pan and zoom.
2. Click **Insert** → **New Map** if you want a second map; otherwise stay on Map.
3. In Catalog, expand `Day1.gdb` (still empty). This is where Part 3 objects will live.
4. Contents pane: turn the basemap on/off. That is a raster-style backdrop; your assets will be vector layers on top.
5. Open **View** → **Catalog View**, then return to the map. Catalog is how you manage data; the map is how you see it.

TIP: After Lab 3, come back here: right-click any feature class → **Attribute Table**. Click one row → it highlights on the map. That is vector data + attributes from the slides.

CHECKPOINT: Student can name Catalog, Contents, Map, and Attribute Table.

5. Part 2 — Coordinate systems

Presentation: Part 2 · Earth, datum, WGS84, GCS, Projected Coordinate System

5.1 What to take from the slides (short)

- Earth is not a perfect sphere. A smooth model is an **ellipsoid**; the gravity surface is the **geoid**.
- A **datum** is the model of Earth that coordinates are measured from. Different datums → same place, different numbers.
- **WGS84** is the usual global datum (GPS, EPSG:4326, degrees).
- **GCS** = latitude / longitude (angles). **PCS** = easting / northing in meters after a projection (example on the slide: **WGS 84 / UTM zone 42N**).
- **Define Projection** only writes the label when coordinates are already correct. **Project** converts coordinates to another CRS.

5.2 Lab 2 — Set the map to the presentation CRS

1. In Contents, right-click the map name (usually **Map**) → **Properties**.
2. Click **Coordinate Systems**.
3. Search `WGS 1984 UTM Zone 42N` (or 41N / 43N if your city is in that zone).
4. Select **WGS 1984 UTM Zone 42N** → **OK**.
5. Bottom of the map view: confirm the CRS name. Move the pointer; values should look like meters (six-digit easting), not 34.5 / 69.2 degrees.
6. Map tab → **Measure** → measure a short street. Units should be meters. That is why the presentation uses a projected system for work, not only GCS.

NOTE: If your training site is not in zone 42N, pick the UTM zone that contains the city. The clicks are the same.

CHECKPOINT: Student can say: GCS = degrees on the ellipsoid; PCS = meters on a flat grid; our map uses UTM.

6. Part 3 — Shapefile, geodatabase, feature dataset, feature classes

Presentation: Introduction to ArcGIS Pro → Managing Vector Data → Geodatabase → Types of Geodatabases → Other GIS Data Types → Feature Dataset

6.1 What to take from the slides (short)

- **Shapefile**: old folder of files (.shp, .dbf, .shx, …). One geometry type. **Cannot** store domains, subtypes, or relationship classes.
- **File geodatabase** (`.gdb`): the container we use in this course.
- **Feature class**: points *or* lines *or* polygons, same fields, same spatial reference.
- **Feature dataset**: a container inside the GDB. All feature classes inside share one coordinate system. Required later for topology / many relationship setups. We put related network layers here.

6.2 Data you will create (no files to import)

Two small models, matching the presentation:

Model	Feature class	Geometry	Primary key	Foreign key (child stores parent's ID)
Irrigation (presentation example)	Scheme	Point	scheme_id	—
	Canal	Polyline	canal_id	scheme_id
	Command_Area	Polygon	area_id	canal_id
	Structure	Point	structure_id	area_id
Telecom (1:M parallel on the irrigation slide)	OLT	Point	olt_id	—
	FDH	Point	fdh_id	olt_id
	Splitter	Point	splitter_id	fdh_id
	Home	Point	home_id	splitter_id
Wireless (subtypes slide)	Tower	Point	tower_id	—

Hierarchy (both 1:M, same idea):

- Irrigation: Scheme → Canal → Command Area → Structure (intake, super passage, weir).
- Telecom: OLT → FDH → Splitter → Home.

6.3 Lab 3A — Create a shapefile (so you see the old format)

1. In Catalog, under your **folder connection** (not inside the .gdb), right-click the folder → **New** → **Shapefile**.
2. Name: `Towers_shp`.
3. Geometry type: **Point**.
4. Coordinate system: same as the map — **WGS 1984 UTM Zone 42N** → Run / Finish.
5. Right-click `Towers_shp` → **Design** → **Fields** (or open Fields view from the ribbon).
6. Add field `tower_id`, type Text, length 20. Save.
7. On disk, look at the folder: you should see several files (`.shp`, `.shx`, `.dbf`, `.prj`). That is a shapefile.

NOTE: We will **not** put domains, subtypes, or relationship classes on this shapefile. Those belong in the geodatabase. Keep the shapefile only as a comparison from the "Other GIS Data Types" slide.

CHECKPOINT: Shapefile exists in the folder; student can name .shp / .dbf / .prj.

6.4 Lab 3B — Create two feature datasets

1. Catalog → expand `Day1.gdb` → right-click the geodatabase → **New** → **Feature Dataset**.
2. Name: `FD_Irrigation`.
3. Coordinate system: **WGS 1984 UTM Zone 42N** (must match the map).
4. Finish.
5. Repeat: New Feature Dataset named `FD_Telecom`, same coordinate system.

CHECKPOINT: `Day1.gdb` shows `FD_Irrigation` and `FD_Telecom`.

6.5 Lab 3C — Create each irrigation feature class

Do these clicks **once per row** in the irrigation table. Example below is for **Scheme**. Then repeat for Canal, Command_Area, Structure.

1. Right-click `FD_Irrigation` → **New** → **Feature Class**.
2. Name: `Scheme`. Alias: Scheme. Geometry: **Point**. Next.
3. Keep the default (do not add z/m unless asked). Next.
4. Fields page — add:
   - `scheme_id` — Text, 20 — this is the **primary key**
   - `scheme_name` — Text, 50
   - `status` — Text, 20 — domain will go here in Lab 4
5. Finish. The feature class appears inside `FD_Irrigation`.

Repeat the wizard for the other irrigation classes. Use this field list:

Feature class	Geometry	Fields to add
Canal	Polyline	`canal_id` (Text 20, PK) · `scheme_id` (Text 20, **FK**) · `canal_name` (Text 50) · `status` (Text 20)
Command_Area	Polygon	`area_id` (Text 20, PK) · `canal_id` (Text 20, **FK**) · `area_name` (Text 50) · `status` (Text 20)
Structure	Point	`structure_id` (Text 20, PK) · `area_id` (Text 20, **FK**) · `struct_type` (Text 30) · `status` (Text 20)

6.6 Lab 3D — Create each telecom feature class

Same wizard, inside `FD_Telecom`:

Feature class	Geometry	Fields to add
OLT	Point	`olt_id` (PK) · `olt_name` · `status`
FDH	Point	`fdh_id` (PK) · `olt_id` (**FK**) · `fdh_name` · `status`
Splitter	Point	`splitter_id` (PK) · `fdh_id` (**FK**) · `split_ratio` (Text 10) · `status`
Home	Point	`home_id` (PK) · `splitter_id` (**FK**) · `address` · `status`
Tower	Point	`tower_id` (PK) · `TowerType` (**Short integer** — subtype field) · `height_m` (Double) · `status`

All ID fields: Text, length 20, unless noted. `TowerType` must be a **short integer** because subtypes use an integer field (presentation: 1 = Macro, 2 = Small cell, 3 = Rooftop).

CHECKPOINT: Nine feature classes exist (4 irrigation + 5 telecom including Tower). Each child class has a foreign-key field named like the parent's ID.

6.7 Lab 3E — Digitize a few features (so keys have values)

You need a handful of points/lines/polygons. Do not import a spreadsheet.

1. Drag `Scheme`, `Canal`, `Command_Area`, `Structure` onto the map.
2. **Edit** tab → **Create**.
3. Select **Scheme** → click **one** point on the map (a project HQ is enough).
4. In the Create Features / Attributes pane, set `scheme_id` = `SCH-01`, name = `North Scheme`, status = `Active`.
5. Select **Canal** → draw **two** lines. For both, set `scheme_id` = `SCH-01`. Use `canal_id` = `CAN-01` and `CAN-02`.
6. Select **Command_Area** → draw **two** polygons. Set `canal_id` to `CAN-01` or `CAN-02`. Use `area_id` = `CA-01`, `CA-02`.

7. Select **Structure** → click **three** points inside the polygons. Set `area_id` to the polygon they sit in. Use types: `intake`, `super passage`, `weir` (presentation list).
8. **Save** edits.
9. Repeat the same idea in `FD_Telecom`: 1 OLT (`OLT-01`), 2 FDHs with `olt_id=OLT-01`, 2 splitters with the matching `fdh_id`, 3 homes with the matching `splitter_id`.
10. Add 3 towers: `tower_id` T-01, T-02, T-03. Leave `TowerType` empty until Lab 4, or type 1, 2, 3.

TIP: The foreign key is just a field you type. `Home.splitter_id` must equal an existing `Splitter.splitter_id`. If the IDs do not match, the relationship class will not find children.

CHECKPOINT: Attribute tables show parent IDs on child rows (1:M in the table, before the relationship class exists).

7. Lab 4 — Domains and subtypes

Presentation: Domains and Subtypes in Geodatabase · Subtypes · example screenshot

7.1 What to take from the slides

- A **domain** lives in the geodatabase. It limits what a field may store (list or min/max).
- A **subtype** splits one feature class into categories using an integer field. Still one table, one geometry type.
- Presentation tower example: `TowerType` 1 = Macro tower, 2 = Small cell, 3 = Rooftop antenna.

7.2 Create a coded-value domain (status list)

1. Catalog → right-click `Day1.gdb` → **Domains**. (Or ribbon: Data → Domains, with the GDB selected.)
2. Click **New**.
3. Domain Name: `Dom_Status`.
4. Description: `Asset status`.
5. Field Type: **Text**.
6. Domain Type: **Coded Value**.
7. In the codes table add:
    - Code `Active` — Description `Active`
    - Code `Planned` — Description `Planned`
    - Code `OnAir` — Description `On Air`
8. On the ribbon, **Save**.

7.3 Create a range domain (tower height)

1. Still in the Domains view, click **New**.
2. Domain Name: `Dom_Height_m`.
3. Field Type: **Double** (or Float).
4. Domain Type: **Range**.
5. Minimum: `3` Maximum: `80`.
6. **Save**. Close the Domains view.

7.4 Assign domains to fields

1. Right-click feature class `Tower` → **Design** → **Fields**.
2. For field `status`, in the **Domain** column choose `Dom_Status`.
3. For field `height_m`, choose `Dom_Height_m`.
4. **Save**.
5. Repeat for every other feature class that has `status`: Scheme, Canal, Command_Area, Structure, OLT, FDH, Splitter, Home. Same domain, assigned once per field.

TIP: After this, editing a `status` cell shows a drop-down instead of free text. That is “rejecting invalid values” from the slide.

7.5 Create subtypes on Tower

1. Right-click `Tower` → **Design** → **Subtypes**.
2. Subtype Field: `TowerType` (the short integer you created).
3. Add codes (presentation values):
   - Code `1` — Description `Macro tower`
   - Code `2` — Description `Small cell`
   - Code `3` — Description `Rooftop antenna`
4. Optional: set a different default `height_m` per subtype (for example Macro 30, Small cell 8, Rooftop 12).
5. **Save**.
6. Start editing Towers. The `TowerType` field should offer Macro / Small cell / Rooftop, not a random integer.

CHECKPOINT: Status is a list; height rejects values outside 3–80; Tower has three subtypes.

8. Lab 5 — Keys and relationship classes

Presentation: Relationship Classes · Example of ERD · Cardinality · Irrigation Example · screenshots

8.1 What to take from the slides

- A relationship class is the geodatabase's foreign-key link.
- **Origin** holds the primary key. **Destination** holds the matching key.
- **1:M** is the usual pattern (one cabinet, many ports; one canal, many command areas; one OLT, many FDHs).
- **Composite**: children belong to the parent and are deleted with it. Use Simple unless you want that behavior.

8.2 Confirm keys before the wizard

Open each child table and check the FK column is populated with real parent IDs from Lab 3E.

Relationship class name	Origin (parent)	Origin PK	Destination (child)	Destination FK	Cardinality
RC_Scheme_Canal	Scheme	scheme_id	Canal	scheme_id	1:M
RC_Canal_CommandArea	Canal	canal_id	Command_Area	canal_id	1:M
RC_CommandArea_Structure	Command_Area	area_id	Structure	area_id	1:M
RC_OLT_FDH	OLT	olt_id	FDH	olt_id	1:M
RC_FDH_Splitter	FDH	fdh_id	Splitter	fdh_id	1:M
RC_Splitter_Home	Splitter	splitter_id	Home	splitter_id	1:M

8.3 Create one relationship class (full clicks)

Example: **Scheme** → **Canal**. You will repeat this for every row in the table.

1. Catalog → right-click `FD_Irrigation` (or the geodatabase) → **New** → **Relationship / Relationship Class**.
2. **Origin table / feature class:** `Scheme`.
3. **Destination table / feature class:** `Canal`.
4. Relationship Class Name: `RC_Scheme_Canal`.
5. Type: **Simple** (not Composite, unless you want canals deleted with the scheme).
6. Cardinality: **One to many (1:M)**.
7. Do not add attributes to the relationship class (Next / No).
8. Origin primary key: `scheme_id`.
9. Origin foreign key (on destination): `scheme_id` on Canal.
10. Forward path label: `feeds canals`. Backward path label: `belongs to scheme`.
11. Finish.

8.4 Repeat for every pair

Run the same wizard for the other five rows. Only the names and key fields change. Always 1:M, Simple, no relationship attributes.

1. Canal → Command_Area   keys: `canal_id` / `canal_id`
2. Command_Area → Structure   keys: `area_id` / `area_id`
3. OLT → FDH   keys: `olt_id` / `olt_id`
4. FDH → Splitter   keys: `fdh_id` / `fdh_id`
5. Splitter → Home   keys: `splitter_id` / `splitter_id`

8.5 Prove it (matches "prove cardinality" without extra data)

1. Add Scheme and Canal to the map if they are not already there.
2. On the map, click the scheme point with **Explore** (or select it).
3. In the pop-up, open the related records / **Canal** link — you should see both canals with `scheme_id = SCH-01`.
4. Alternatively: select the scheme → **Map** tab → **Select by Relationship** (or right-click layer → Selection → Select Related Records, depending on Pro version).
5. Do the same for one splitter: related homes should be the three houses you typed with that `splitter_id`.

CHECKPOINT: Six relationship classes in the GDB. Selecting one parent highlights or lists its children. Student can say where the FK is stored (on the child).

NOTE: Tower has no child class in this manual. Subtypes already split towers inside one feature class. If you need Tower → Sector later, add a `Sector` table with `tower_id` and create `RC_Tower_Sector` the same way.

9. Lab 6 — Raster analysis and DEM

Presentation: Part 4 · Raster Analysis · Weighted Overlay · Resolution · DEM · DEM sources · What can be extracted · Let's Explore It in ArcGIS Pro

9.1 What to take from the slides (short)

- Rasters store continuous surfaces (elevation, rain, imagery).
- Common tools: Reclassify, Euclidean Distance, Raster Calculator, Viewshed, **Weighted Overlay** (combine scored layers with weights).
- Resolution = cell size. A 30 m DEM is one elevation for each 30 × 30 m square.
- From a DEM you can derive slope, aspect, hillshade, contours, viewshed, watershed, elevation profile.

9.2 Add a DEM (use any real DEM you already have)

Do not build a fake elevation table. Use one of these, in this order of ease:

1. **Living Atlas / ArcGIS Online:** Catalog → Portal → Living Atlas → search `Terrain` or `DEM` → add a world elevation or local DEM to the map.
2. **Local file:** if the class has an SRTM/ALOS GeoTIFF, Catalog → right-click the folder → Add to map.
3. Right-click the DEM in Contents → **Properties** → Source. Write down cell size (resolution) and CRS. Project it to UTM if it is still in WGS84 degrees and you will run Slope.

9.3 Extract what the presentation lists

Analysis tab → **Tools**. Search and run each tool. Input raster = your DEM. Output goes into `Day1.gdb`.

Tool	Geoprocessing tool name	Why (from the slide)
Slope	Slope (Spatial Analyst)	Steepness for access, erosion, routing
Aspect	Aspect	Direction the slope faces
Hillshade	Hillshade	Shaded relief for map reading
Contours	Contour	Lines of equal elevation (try 20 m interval)
Viewshed	Viewshed or Geodesic Viewshed	What is visible from a Tower point

1. Run **Hillshade** first so the class sees terrain immediately.
2. Run **Slope**. Symbolize with a simple yellow–red ramp.
3. Optional if Spatial Analyst is licensed: **Weighted Overlay** — reclassify slope to 1–5, give it 100% weight as a one-layer demo, or add a second raster if you have one. Do not invent extra criteria tables.
4. Optional: **Viewshed** with one Tower as the observer, offset 30 m, to match the slide.

NOTE: These tools need the Spatial Analyst extension (and 3D Analyst for some visibility tools). If the extension is off: Project → Licensing, or skip to Hillshade if it is available as a raster function: Contents → raster layer → Raster Functions → Hillshade.

CHECKPOINT: DEM on the map; at least Hillshade and Slope created. Student can state cell size in meters.

10. Expected outputs

#	Deliverable	Done?
1	Project `Day1_GIS_Telecom.aprx` and `Day1.gdb`	

2	Map CRS = WGS 1984 UTM (zone for your city)	
3	Shapefile `Towers_shp` in the folder (comparison only)	
4	Feature datasets `FD_Irrigation` and `FD_Telecom`	
5	All feature classes from the schema tables, with PK and FK fields	
6	Domains `Dom_Status` and `Dom_Height_m` assigned	
7	Subtypes on Tower (Macro / Small cell / Rooftop)	
8	Six 1:M relationship classes; related records work	
9	DEM + Hillshade + Slope	

11. Quick quiz (same ideas as the slides)

1. Name the five GIS components from the slide.
2. Vector or raster for a fiber cable? For a DEM?
3. What is the difference between a datum and a projected coordinate system?
4. Why can a shapefile not hold a relationship class?
5. Where is the foreign key stored in a 1:M (Splitter → Home)?
6. What integer codes did we use for Tower subtypes?
7. Name three rasters you can derive from a DEM.

Answers:

1. Hardware, software, data, people, methods/procedures.
2. Fiber = vector line. DEM = raster.
3. Datum = Earth model for coordinates. PCS = flat meter grid created by projecting a GCS.
4. Shapefiles have no geodatabase behavior (no domains, subtypes, relationship classes).
5. On the child: `Home.splitter_id`.
6. 1 Macro, 2 Small cell, 3 Rooftop.
7. Any three of: slope, aspect, hillshade, contour, viewshed, watershed, profile.

12. Instructor notes

- Teach from **Day 1_ready.pptx**. After each part-divider slide, switch to the matching lab.
- Delete presentation slide 51 (yellow correction log) before class.
- If Pro's New Relationship Class wizard labels differ slightly by version, the meaning is the same: origin, destination, 1:M, origin PK, destination FK.
- Do not distribute the old Day 1 CSV/GeoJSON pack for this version. It is not needed.

13. Version history

Version	Changes
v3	MetroTel synthetic datasets and concept-heavy labs (previous)
v4	Aligned to Day 1_ready.pptx. No synthetic tables. Click-by-click: shapefile, feature dataset, feature classes, domains, subtypes, keys, relationship classes, DEM derivatives.

— End of Day 1 Manual (Version 4.0) —